

11. Status of Higgs Boson Physics

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11.1 Introduction

Understanding how the masses of the known elementary particles are generated has been one of the fundamental endeavours in particle physics for several decades. The chiral nature of the weak interactions forbids an intrinsic mass terms for the quarks and leptons at high energy, and their

with $\phi^0 = H + \langle \phi^0 \rangle$ and $\langle \phi^0 \rangle \equiv v$, inducing the spontaneous breaking of the SM gauge symmetry $SU(3)_C \times SU(2)_L \times U(1)_Y$ into $SU(3)_C \times U(1)_{\text{em}}$. The global minimum of the theory defines the ground state, and spontaneous symmetry breaking implies that there is a (global and/or local) symmetry of the system that is not respected by the ground state. From the four generators of the $SU(2)_L \times U(1)_Y$ SM gauge group, three are spontaneously broken, i.e., they do not leave the ground state invariant. This breaking implies the existence of three massless Goldstone bosons identified with three of the four Higgs field degrees of freedom. The Higgs field couples to the W_μ and B_μ gauge fields associated with the $SU(2)_L \times U(1)_Y$ local symmetry through the covariant derivative appearing in the kinetic term of the Higgs Lagrangian,

$$\mathcal{L}_{\text{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \quad (11.4)$$

where $D_\mu \Phi = (\partial_\mu + ig\sigma^a W_\mu^a/2 + ig'Y B_\mu/2)\Phi$, g and g' are the $SU(2)_L$ and $U(1)_Y$ gauge couplings, respectively, and $\sigma^a, a = 1, 2, 3$ are the usual Pauli matrices. As a result, the neutral and the two charged massless Goldstone degrees of freedom mix with the gauge fields corresponding to the broken generators of $SU(2)_L \times U(1)_Y$ and become, in the unitarity gauge, the longitudinal components of the Z and W physical gauge bosons, respectively. The Z and W gauge bosons acquire masses,

$$m_W^2 = \frac{g^2 v^2}{4}, \quad m_Z^2 = \frac{(g'^2 + g^2)v^2}{4}. \quad (11.5)$$

The fourth generator remains unbroken since it is the one associated to the conserved $U(1)_{\text{em}}$ gauge symmetry, and its corresponding gauge field, the photon, remains massless. Similarly, the eight color gauge bosons, the gluons, corresponding to the conserved $SU(3)_C$ gauge symmetry with 8 unbroken generators, also remain massless (though confined inside hadrons and mesons as the result of the asymptotic freedom behaviour of Quantum Chromodynamics (QCD)). Hence, from the initial four degrees of freedom of the Higgs field, two are absorbed by the $W^\pm$ gauge bosons, one by the Z gauge boson, and there is one remaining degree of freedom, H , that is the physical Higgs boson — a new scalar particle first imagined by P. Higgs [4]. The Higgs boson is neutral under the electromagnetic interactions and transforms as a singlet under $SU(3)_C$ and hence does not couple at tree level to the massless photons and gluons.

The fermions of the SM acquire a mass through renormalisable and gauge invariant interactions between the Higgs field and the fermions: the Yukawa interactions,

$$\mathcal{L}_{\text{Yukawa}} = -\hat{h}_{d_{ij}} \bar{q}_{L_i} \Phi d_{R_j} - \hat{h}_{u_{ij}} \bar{q}_{L_i} \tilde{\Phi} u_{R_j} - \hat{h}_{l_{ij}} \bar{l}_{L_i} \Phi e_{R_j} + h.c., \quad (11.6)$$

which respect the symmetries of the SM but generate fermion masses once EWSB occurs. In the Lagrangian above, $\tilde{\Phi} = i\sigma_2 \Phi^*$ and q_L (l_L) and u_R , d_R (e_R) are the quark (lepton) $SU(2)_L$ doublets and singlets, respectively, while in each term, $\hat{h}_{f_{ij}}$, $f = u, d, l$ is parametrised by a 3×3 matrix in family space. The mass term for neutrinos is omitted, but could be added in an analogous manner to the up-type quarks when right-handed neutrinos are supplementing the SM particle content (neutrinos can also acquire Majorana masses via non-renormalisable dimension-5 interactions with the Higgs field [19]). Once the Higgs field acquires a VEV, and after rotation to the fermion mass eigenstate basis that also diagonalises the Higgs-fermion interactions, $\hat{h}_{f_{ij}} \rightarrow h_{f_i} \delta_{ij}$ (the diagonal Yukawa coupling h_{f_i} is often denoted as y_{f_i}), all fermions acquire a mass given by $m_{f_i} = h_{f_i} v / \sqrt{2}$. The indices $i, j = 1, 2, 3$ refer to the three families in the up-quark, down-quark or charged lepton sectors. Remarkably, if the Yukawa interactions Eq. (11.6) are indeed fully responsible for the fermion masses, the Higgs interactions do not mediate flavour changing neutral currents at tree-level. It should be further noted that the EWSB mechanism provides no additional insight on possible underlying reasons for the large variety of mass values of the fermions, often referred to as